

E-commerce Sales Investigation

01_create_tables

Synthetic Dataset for the project using Chatgpt.com:

1. Orders_raw.csv - 507
2. Customers_raw.csv - 30
3. Products_raw.csv - 5015
4. Payments_raw.csv - 5333

Loaded tables into mysql workbench:

1. Orders_raw.csv - loaded directly through 'table data import wizard'
2. Customers_raw.csv - tried loading directly but all records did not get imported. So, created a table, added columns, data type, not null status and then loaded data again into the existing table through 'table data import wizard'
3. Products_raw.csv - tried loading directly but all records did not get imported. So, created a table, added columns, data type, not null status and then loaded data again into the existing table through 'table data import wizard'
4. Payments_raw.csv - tried loading directly but all records did not get imported. So, created a table, added columns, data type, not null status and then loaded data again into the existing table through 'table data import wizard'

Checked the existence of table with `SELECT * FROM table_name;`

02_validate_customers

1. Checked row count - 507
2. Preview of data - to get a display of first 10 rows
3. Description of customers_raw table -

Field	Type	Null	Key	Default	Extra
state	text	YES			
signup_date	text	YES			
Region	text	YES			
email	text	YES			
customer_segment	text	YES			
customer_name	text	YES			
customer_id	int	YES			
city	text	YES			
4. Retrieve NULL/blank values from the customers_raw table:
 - Missing_customer_id - 0
 - Missing_name - 0
 - Missing_email - 8
 - Missing_city - 4
 - Missing_state - 4
 - Missing_region - 4

- Missing_signup_date - 0
 - Missing_segment - 0
5. Duplicate primary key records - 7
 6. Duplicate customer records (name, email) - 7
 7. Distinct values/categorical validation:

- Region

Region	COUNT(*)
West	133
North	116
South	115
East	68
Central	66
West	4
blank	4
W	1

- City

City	COUNT(*)
Nagpur	38
Bhopal	34
Kochi	33
Indore	31
Bhubaneswar	30
Hyderabad	30
Mumbai	30
Jaipur	28
Chennai	28
Delhi	27
Surat	24
Pune	23
Kolkata	21
Bengaluru	21
Lucknow	20
Noida	20
Patna	20
Gurugram	20
Ahmedabad	19
blank	4
Pune	2
Bangalore	2
New Delhi	2

- State

State	COUNT(*)
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Maharashtra	90
Madhya Pradesh	65
Gujarat	43
Uttar Pradesh	40
Kerala	33
Odisha	30
Telangana	30
Delhi	29
Tamil Nadu	28
Rajasthan	27
Karnataka	23
Haryana	22
West Bengal	21
Bihar	20
blank	4
MH	2

- customer_segment
Customer_segment | COUNT(*)
Consumer 401
Small Business 83
Corporate 23

8. White space issues - 21 records with respect to customer_name, email, city, state, region
9. Validating email inconsistencies - 13 records
10. Signup_date greater than current_date has 0 records

04 validate products

1. Row count - 30
2. Preview of data
3. Description of data
4. NULL/blank data: missing_subcat = 3, missing_brand = 3, missing_unit_cost = 3, missing_supplier_id = 2
5. 0 duplicate primary key
6. 0 duplicate product records
7. Distinct values for categorical validation
 - Product_name: no inconsistencies
 - Category: no inconsistencies
 - Subcategory: 3 blanks
 - Brand: 3 blanks
 - Product_status: no inconsistencies
8. White space issues for categorical columns: 3 records
9. Numeric validity
 - Unit_cost < 0: no inconsistencies

- List_price < 0: no inconsistencies
- Unit_cost > list_price: 11 records

06 validate payments

1. row count: 5333
2. preview of data
3. description of data
4. NULL/blank data
 - Missing_payment_date: 30 records
 - Missing_amount_paid: 18 records
 - missing_transaction_id : 25 records
5. duplication of payments: payment_id (which is not primary key), transaction_id: 5 records
6. distinct values for categorical records like
 - Payment_id: 5 records are duplicated
 - Order_id: multiple duplicated records
 - Payment_method: UPI, Credit Card, Cash on Delivery, Net Banking, Debit Card, PayPal, COD, UPI, CC
 - Payment_status: no inconsistencies
 - Transaction_id: 25 blank records, 1 transaction_id with 3 records, 13 transaction_id with 2 records
7. numeric validity:
 - if amount_paid < 0 and payment_status <> 'Refunded': 5
 - if amount_paid == 0 and payment_status <> 'Failed': 20
 - if amount_paid > 0 and payment_status IS 'Refunded':6
 - if amount_paid > 0 and payment_status <> 'Failed':4
 - if payment_method IS NOT 'Credit Card' and payment_status <> 'Pending':236
8. date validation: payment_date < current_date(): 0
9. referential integrity: if order_id in payments_raw exists in orders_raw: 12
10. White space inconsistencies across categorical columns: 12

08 validate orders

1. Row count: 5015
2. preview of data
3. description of data
4. NULL/blank records
 - Missing_customer_id: 25
 - Missing_shipping_date: 74
 - missing_price:35
5. duplication of orders: order_id (not primary key), product_id, quantity, price: 15 records
6. distinct values for categorical columns
 - Order_id: 15
 - Region: West, North, South, East, Central, South, West, W

- Payment_method: UPI, Credit Card, Cash on Delivery, Debit Card, Net Banking, PayPal, COD, UPI, CC
 - Order_status: no inconsistencies
 - Sales_channel: no inconsistencies
7. numeric validation:
 - quantity <= 0: 11
 - price < 0: 0
 - discount < 0 or discount > 1: 28
 - shipping_cost < 0: 0
 8. date validation
 - if order_date > current_date():0
 - if shipping_date > current_date():0
 - if order_date < shipping_date: 196
 9. referential integrity
 - customer_id exists in customers_raw: 37
 - product_id exists in products_raw: 10
 10. business rule
 - order_status = 'Cancelled' AND shipping_date IS NOT NULL: 337
 - order_status = 'Pending' and shipping_date is not null: 307
 11. region consistency - compare orders_raw.region with customers_raw.region for the same customer: 208
 12. white space inconsistencies for categorical columns: 45 records

03 clean customers

1. create cleaned table : created customers_cleaned table and inserted data from customers_raw into it
2. Verified with count if all rows are transferred
3. standardize text formatting: changed all columns and rows to lowercase and used to trim to trim first and last white spaces
4. handle blank values: found blanks in
 - Blank_email: 8
 - Blank_city: 4
 - Blank_state: 4
 - Blank_region: 4

MySQL threw error 1175. So, safe update mode was disabled using
SET SQL_SAFE_UPDATES = 0;
5. convert empty strings to NULL: converted all blank strings to NULL successfully
6. standardize known categorical inconsistencies: Identified unique records and standardized them
7. Region: 6 (North, West, East, Central, South, NULL)
8. State: 14
9. standardize city names: 20
10. check and clean email inconsistencies: 5 records changed manually
11. investigate and handle duplicate customers

12. DO NOT DELETE DUPLICATES BLINDLY

13. review all records first: all 7 records are identical. Created another table where all the unique records are transferred. Dropped customers_cleaned. Renamed that table again as customers_cleaned.
14. standardize signup_date to date data type: added column temp_date with date data type. Copied records from signup_date to temp_date with str_to_date() function. Had to use SET SQL_SAFE_UPDATES = 0. Dropped the column signup_date. Renamed temp_date as signup_date.
15. validate cleaned table: all checks done
16. export data to csv: customers_cleaned.csv exported

05 clean_products

1. Create cleaned table: created products_cleaned like products_raw and inserted data from products_raw into it. Verified by the data and number of rows.
2. Standardize text - TRIM product_name - TRIM category - TRIM subcategory - TRIM brand - TRIM product_status - Convert categorical/text values to lowercase
3. Investigate unit_cost > list_price
 - Review the 11 records: found 2 records
 - Correct only if there is a clear reason, otherwise decide how to flag/handle them: adding a column flag and mentioning the reason 'unit_cost > list_price'
4. Subcategories with multiple categories: audio, computer accessories, mobile accessories, stationery. Added a category_flag column for product_name: ball pen pack, usb hub, power bank with 'category mismatch' record
5. Handle NULL/blank values - subcategory → investigate and decide how to handle - brand → investigate and decide how to handle - supplier_id → handle missing values - unit_cost → investigate missing values
 - Blank_subcategory: 3, blank_brand: 3, blank_supplier_id: 2
 - For blank subcategories, updated those as 'skincare', 'kitchen appliances', 'home office'
 - For blank brands, added 'generic' to the column.
 - For blank supplier_id: added another column 'supplier_id_flag' and mentioned them as 'missing_supplier_id'
6. Change in data type from varchar to int for the column 'supplier_id': removed NOT NULL constraint, checked if there are any other characters than numerics, altered the column to int data type

07 clean_orders

1. Created payments_cleaned table like payments_raw, matched the row count, matched the data type
2. Updated the payments_cleaned such that all categorical/text columns and converted categorical values to lowercase
3. Standardize payment methods
 - cod → cash on delivery: updated and verified the count to be 0

- cc → credit card: updated and verified the count to be 0
- 4. Handle missing values
 - customer_id → investigate 25
 - shipping_date → investigate 74
 - price → investigate 35
- 5. Investigate duplicate order records
 - Review the 15 duplicate (order_id + product_id + quantity + price)
 - Remove only confirmed duplicates
- 6. Handle invalid numeric values
 - quantity ≤ 0 → investigate/correct/remove
 - discount < 0 or > 1 → investigate/correct
 - price < 0 → none found
 - shipping_cost < 0 → none found
- 7. Handle date inconsistencies
 - order_date $>$ current date → none
 - shipping_date $>$ current date → none
 - shipping_date before order_date → investigate 196
- 8. Handle broken relationships
 - invalid/missing customer_id → investigate 37
 - invalid product_id → investigate 10
- 9. Handle business-rule inconsistencies
 - Cancelled + shipping_date → investigate 337
 - Pending + shipping_date → investigate 307
- 10. Handle region inconsistencies
 - Compare order region with customer region
 - Investigate 208 mismatches
- 11. Re-validate orders_cleaned

07 clean payments

1. Created payments_cleaned table like payments_raw, matched the row count, matched the data type
2. Updated the payments_cleaned such that all categorical/text columns and converted categorical values to lowercase
3. Standardize payment methods
 - cod → cash on delivery: updated and verified the count to be 0
 - cc → credit card: updated and verified the count to be 0
4. Handle missing values
 - payment_date → investigate/handle 30: for payment_status = pending and failed, payment_date is null. For payment_status = paid and refunded, payment_date_flag column was added and the records were 'missing_payment_date'

- amount_paid → investigate/handle 18: for payment_status = pending and failed, payment_date is null. For payment_status = paid, payment_date_flag column was added and the records were 'missing_payment_date'

- transaction_id → investigate/handle 25

5. Investigate duplicate payment_id / transaction_id

- Review the 5 duplicated records
- Don't automatically delete them

6. Investigate invalid order_ids

- Review the 12 records that don't match orders_raw
- Decide whether to remove, correct, or flag them

7. Investigate payment amount/status inconsistencies

- negative amount + non-refunded
- zero amount + non-failed
- positive amount + refunded
- positive amount + failed

8. Re-validate payments_cleaned